

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 4

Calculus - Integration & Applications

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find $\int (3x^2 + 2x) dx$ [2 marks]

2. Evaluate $\int_0^2 (x + 1) dx$ [3 marks]

3. Find $\int \cos x \, dx$ [2 marks]

4. Find $\int 1/x \, dx$ [2 marks]

5. Evaluate $\int_0^{\pi} \sin x \, dx$ [3 marks]

6. Find $\int e^x \, dx$ [2 marks]

7. Find the area under $y = x$ from $x = 0$ to $x = 3$ [3 marks]

8. Integrate: $\int (x - 1)^2 \, dx$ [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Find the area between $y = x^2$ and $y = 4$ from $x = 0$ to $x = 2$. (6 marks)

10. Evaluate $\int_1^3 (1/x^2) dx$. (3 marks)

11. A curve passes through $(0, 2)$ with gradient $2x + 1$. Find equation of curve. (5 marks)

12. Find the volume of solid generated by rotating $y = \sqrt{x}$ about x-axis from $x = 0$ to $x = 4$. (6 marks)

13. Evaluate $\int_0^{\pi/2} \cos 2x \, dx$. (4 marks)

14. The velocity of object is $v = 3t + 2$. If at $t = 0$, $s = 5$, find displacement at $t = 3$. (5 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Integrate $3x^2 + 2x$

$$\int (3x^2 + 2x) dx = x^3 + x^2 + C$$

Q2: Evaluate $\int_{-2}^2 (x+1) dx$

$$[x^2/2 + x]_{-2}^2 = 2 + 2 - 0 = 4$$

Q3: Integrate $\cos x$

$$\int \cos x \, dx = \sin x + C$$

Q4: Integrate $1/x$

$$\int (1/x) dx = \ln|x| + C$$

Q5: Evaluate $\int_0^\pi \sin x \, dx$

$$[-\cos x]_0^\pi = -\cos \pi + \cos 0 = 1 + 1 = 2$$

Q6: Integrate e^x

$$\int e^x \, dx = e^x + C$$

Q7: Area under $y = x$ from 0 to 3

$$\text{Area} = [x^2/2]_0^3 = 9/2 = 4.5 \text{ square units}$$

Q8: Integrate $(x-1)^2$

$$= \int (x^2 - 2x + 1) dx = x^3/3 - x^2 + x + C$$

SECTION B SOLUTIONS

Q9: Area between curves

$$\text{Area} = \int_0^2 (4 - x^2) dx = [4x - x^3/3]_0^2 = 8 - 8/3 = 16/3 \text{ square units}$$

Q10: Evaluate $\int_1^3 (1/x^2) dx$

$$[-1/x]_1^3 = -1/3 - (-1) = 2/3$$

Q11: Find equation of curve

$$dy/dx = 2x + 1, y = \int (2x + 1) dx = x^2 + x + C, \text{ At } (0,2): 2 = 0 + 0 + C, C = 2, y = x^2 + x + 2$$

Q12: Volume of revolution

$$V = \pi \int_0^1 (\sqrt{x})^2 dx = \pi \int_0^1 x \, dx = \pi [x^2/2]_0^1 = 8\pi \text{ cubic units}$$

Q13: Evaluate $\int_0^{\pi/2} \cos 2x \, dx$

$$[\sin 2x/2]_0^{\pi/2} = \sin \pi/2 - \sin 0/2 = 1/2$$

Q14: Find displacement

$$s = \int (3t + 2) dt = 1.5t^2 + 2t + C, \text{ At } t=0, s=5: C=5, s = 1.5t^2 + 2t + 5, \text{ At } t=3: s = 13.5 + 6 + 5 = 24.5$$